RI080

Typed Judgment, Deterministic Enforcement: A Consent-Gated Gateway Architecture for Inter-Agent Collaboration over Messaging Channels

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Introduction

Messaging channels deliver human events that are unstructured, duplicated, delayed, or replayed, while agent protocols expect authenticated endpoints, structured requests, and explicit task state. A gateway must decide whether a message requests work, which agent is eligible, whether consent is needed, and whether an event is a replay or an injection.

We separate bounded model judgment from deterministic enforcement: the model answers seven closed questions (J1 intent, J2 routing, J3 ambiguity, J4 consent, J5 replay, J6 retryability, J7 injection) and policy code owns thresholds and every effect.

Objectives

- Seven modules from an iMessage/Photon ingress to an ACPs/AIP v2.2 Leader–Partner path, with four invariants kept in code: authorization, identity, isolation, idempotency.
- A judgment plane of seven closed questions that returns labels or probabilities only and cannot emit free text or call a tool.
- Live measurement of routing, consent, the advisory judgment path, and the gateway's own handling cost.
- A released artifact that reproduces the request suites, the paired ablations and the cached judgment responses.

Materials and Methods

Modules. M1 ingress authentication; M2 caller/conversation isolation; M3 discovery and routing; M4 consent; M5 identity-bound execution; M6 state, idempotency and recovery; M7 delivery and telemetry.

Evaluation. 130 generated requests through the real ingress, three deliveries each; paired same-code ablations on staging copies; 30 consent trials plus 10 redeliveries on the deployed adapter; a 280-item judgment suite sampled three times; a 456-item injection stress set.

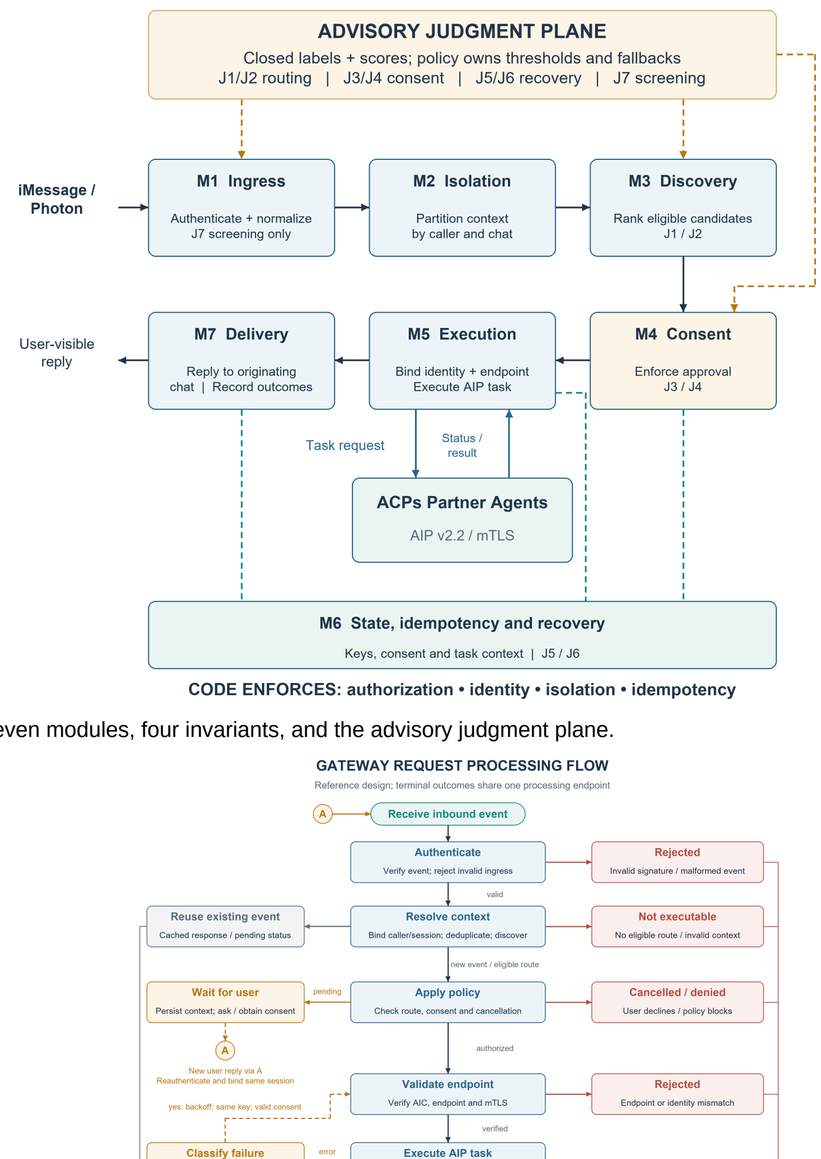


Figure 1. Seven modules, four invariants, and the advisory judgment plane.

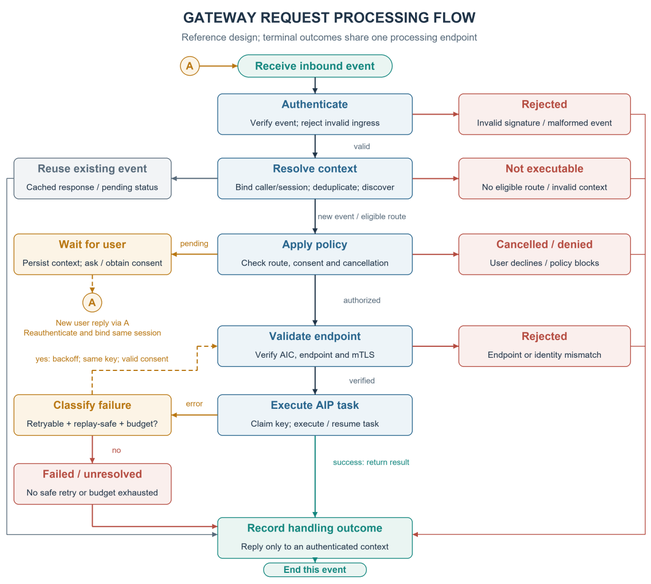


Figure 2. Deduplication precedes every model call; a pending consent re-enters at authentication; a failed identity or endpoint check never reaches execution.

Results

Routing. The generated target was recommended for 48/48/50 of 50 direct requests; 39/38/38 of 40 ambiguous requests received more than one agent; 0/40 out-of-scope requests recommended a study partner.

Ablation	Control	Removed
Identity binding	0/30	30/30
Caller isolation	0/30	30/30
Idempotency	30 exec.	60 exec.

Consent. No partner action before consent (0/40); every consent produced exactly one action (36/36).

	Typed	Free text
Accuracy	94.1%	83.9%
Brier	0.095	0.266
ECE	0.026	0.144

The injection stress set (200 injected, 256 benign near-misses) misses none but flags about half of quoted mentions, which is why the injection score only parameterizes policy.

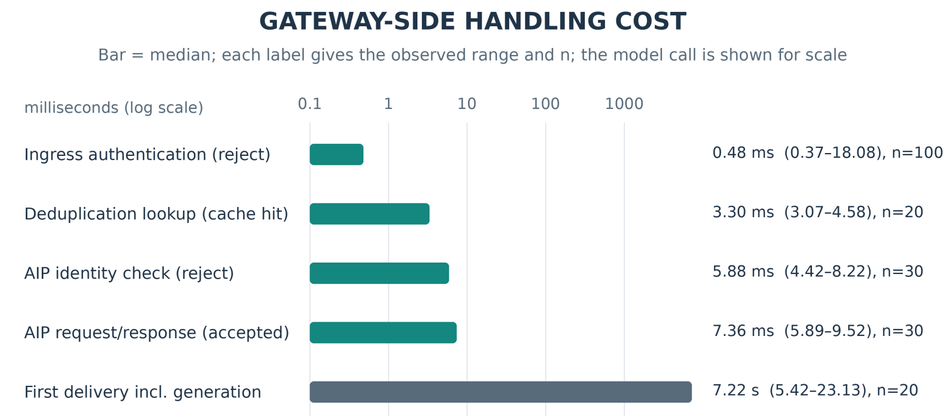


Figure 3. Median gateway-side handling cost per operation.

Conclusion

The model advises; code controls identity, state, and every side effect. Typed answers are more accurate and better calibrated than free text, and consent gating admits exactly one partner action per accepted event. At a glance: 48/48/50 targets recommended, 0/30 to 30/30 under the identity ablation, 36/36 one-action consents, and 94.1% versus 83.9% accuracy.

References

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Artifact: github.com/ankelas/consent-gated-gateway-artifact

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